NMMA Codebase Review — Retreat Prep

Generated by Claude Code

2026-08-23 *Research only — no code changed, nothing pushed/merged*

Scope: `nmma/` source tree, tests, docs, CI/lint config, and the live GitHub issue/PR tracker for `nuclear-multimessenger-astro`
Goal: give the team a prioritized, evidence-based punch list for a 1-week retreat, split by who should do the work. Findings are checked directly against the source; a few counts (e.g. the number of `FIXME` comments) are approximate since the exact number depends on the grep pattern used — treat them as order-of-magnitude, not exact.

Note: findings below come from static grep/read/`git log` inspection rather than a live `flake8` run (`healpy` and `flake8` are not available in the environment used for this analysis). This is solid for locating problems, but running `flake8 -max-complexity 15` in the actual `nmma` dev env (`pip install -e ".[dev,grb,neuralnet]"`) should be the first thing done at the retreat to corroborate and extend §4.2.

1 Headline findings

1. **Five of eight top-level modules have essentially zero dedicated tests**: `eos/` (1,775 lines), `gw/` (311 lines), `mlmodel/` (1,333 lines), `population/` (28 lines), `post_processing/` (1,425 lines, only touched incidentally via `maximum_mass.py`). That's roughly **4,850 lines of core physics/inference code with no correctness test**. `nmma/core/mpi_setup.py` (683 lines, the MPI-parallel `pbilby_sampling` path used whenever `USE_MPI and sampler == 'dynesty'`) also has zero coverage — exactly the kind of ordering-sensitive distributed code that regresses silently. The maintainers already know part of this — issue **#196** **"Tests for nmma/eos"** and **#299** **"Joint analysis unit test"** have been open since **2023-08** and **2024-01** respectively.
2. **`nmma/tests/models.py` is entirely `pytest.mark.skip`'d** (GitLab SVD download path retired) and never replaced with an equivalent test — a whole test file is dead weight that silently reports 0 real coverage while looking like it exists. Separately, **test collection breaks for 4 of 14 test modules when `healpy` isn't installed** (`maximum_mass.py`, `systematics.py`, `tools.py`, `training.py` all fail with `ModuleNotFoundError: No module named 'healpy'`) — this traces back to `nmma/em/__init__.py` eagerly importing the entire `em` submodule tree, so even a small, unrelated import drags in `healpy` and other heavy/optional dependencies. The failure is environment-dependent, but the import coupling itself is real and worth loosening regardless.
3. **Two concrete, currently-shipping bugs, both cheap to fix and both untested:**
   - `nmma/core/base.py`'s `constraints` setter (∼lines 55–65): if `value` isn't a `PriorDict`, `Constraint`, or `dict`, the local `constr` variable is never assigned, and `self._constraints = constr` raises an opaque `UnboundLocalError` instead of a clear validation error.
   - `nmma/population/pop_likelihood.py`'s `NeutronStarPopulation.__init__` only handles `model_name in {'flat', 'peak'}`; any other value silently leaves `self.distribution` unset, surfacing later as a confusing `AttributeError` in `log_likelihood()` rather than a clear error at construction time.
4. ∼**50–65 `FIXME/TODO` comments** (exact count depends on the grep pattern used — treat as approximate), many self-flagged as broken or approximate: `nmma/em/lightcurve_generation.py:145` questions its own redshift-correction exponent; `nmma/core/conversion.py:549` flags a physics inconsistency between BNS and NSBH GRB channels ("NSBH might produce a GRB too, why not provide the same expression for BNS?"); `nmma/em/io.py:545` is tagged "Legacy??? seems unused." Densest clusters: `nmma/joint/injection_handling.py` (11, including a self-flagged fragile dependency on undocumented bilby internals at line 300) and `nmma/em/model.py` (10, including *"This is an unclean default, should be more explicit!"* and *"This is incomplete, but identical to handling in NMMA 0.2.2"* — i.e. known-incomplete logic shipped on purpose to match old behavior). These are landmines: they read as "someone

already knows this is wrong," but nobody has turned them into tracked issues or tests.

5. **11 bare `except:` clauses** across `core/`, `em/`, `eos/`, `post_processing/`: `nmma/eos/eos_likelihood.py:137,154`, `nmma/eos/eos_processing.py:57`, `nmma/core/base.py:246,379`, `nmma/core/utils.py:21`, `nmma/core/mpi_setup.py` `nmma/em/training.py:532,584`, `nmma/post_processing/maximum_mass_constraint.py:159`, `nmma/post_processing` — these swallow `KeyboardInterrupt`/`SystemExit` and make debugging silent failures in long sampler runs much harder. Related: **106 `print()` calls in production code**, inconsistently mixed with real logging (e.g. `nmma/core/base.py` uses `logger.info` in one function and plain `print()` in `bilby_sampling` at lines 297–298, 308) — no consistent logging discipline.

6. **`nmma/mlmodel/` is the most stale, least tested subsystem in the repo.** Untouched since 2024-07-18 (`inference.py` since 2025-01-09) while the rest of core/em/joint/eos were touched as recently as June–July 2026; zero tests; ships 9.6 MB of binary model weights (`frozen-flow-weights.pth` 8.0 MB, `similarity_embedding_weights.pth` 1.6 MB) committed directly to git — not via LFS, not covered by any `.gitattributes` binary rule — added in 2024 and never revisited. Needs an explicit decision: maintain it (add tests, document it) or mark it clearly experimental/deprecated.

7. **Docs have two actively-broken pages, plus thin docstrings.** `doc/index.rst` (the Sphinx "Installation" page) and `doc/lfi_analysis.md:20` tell new users to `pip install -r requirements.txt` / `pip install -r ml_requirments.txt` — **neither file exists in the repo anymore**; the project moved to `pyproject.toml` extras (`pip install -e ".[dev,grb,neuralnet]"`, per CLAUDE.md) and the docs were never updated. Separately, `doc/quick-start-guide.rst` (untouched since 2023-09-19) documents CLI flags `-aligned-spin` and `-injection-outfile` that **do not exist anywhere in the source tree** — copy-pasting its example command fails with an argparse error today. A brand-new user following either page verbatim hits an error on step 1. Beyond that: `nmma/core/base.py` (436 lines, the shared likelihood base every domain extends) has ∼4 docstrings; `nmma/joint/joint_likelihood.py` has ∼3; `nmma/em/model.py` (1720 lines, 30+ model classes) is better but still partial (∼27 docstrings). `tutorials/` has only 2 notebooks against 25+ documented models and ∼17 CLI entry points. `README.md` (166 lines) has no installation instructions, CLI quick-start, or code examples at all — citation/funding content only. `doc/models.md` links to a stale line-anchor in `nmma/em/model.py` (points at line 72; actual content is now ∼97–125) — model names themselves are still accurate, only the link target drifted. Further drift: issues **#374**, **#307**, **#213**/**#136**, and doc pages tied to the still-open **#388** design question.

8. **Tooling/CI pins are stale and inconsistent.** `.pre-commit-config.yaml` pins `black==22.3.0` (current stable is 24.x/25.x) and `pre-commit-hooks==v3.3.0` (current is v5.x). `.flake8` globally ignores `E501` with `max-line-length = 80` set but never enforced — decorative config. `.github/workflows/*.yml` mixes `actions/checkout@v2`/`@v3`/`@v4`, and `publish-to-pypi.yml:13` uses `actions/checkout@main` (tracks a moving branch — a supply-chain risk specifically in the *publish* workflow), plus `actions/setup-python@v3` (current is v5).

9. **PR queue has two multi-megabyte-diff PRs** (#105, #82: +3.3M lines each) sitting open since June 2023 — almost certainly stale branches with binary/pickle files being diffed as text (no `.gitattributes` rule stops this). A `.gitattributes` rule (e.g. `*.pkl -diff -text`, `*.pth -diff -text`) would prevent recurrence.

10. **Trivial repo hygiene:** `doc/_static/nmma-docs.css~` and `doc/_static/.nmma-docs.css.un~` are vim backup/swap files checked into git since 2022. `nmma/core/utils.py:129` emits a live `DeprecationWarning` from an un-prefixed regex string (`sep='\s+'` should be `sep=r'\s+'`). `loadEventSpec()` in `nmma/em/io.py:546` is confirmed dead code (already self-flagged, zero references elsewhere).

11. **Design-principle read: the core abstraction is genuinely sound; the joint pipeline's glue code is the weak point.** `NMMALikelihoodMixin`/`NMMALikelihood` (`nmma/core/base.py:37-185`) cleanly define the `parameter_conversion`/`sub_log_likelihood`/`sanity_checks`/ `constraints` contract, extended uniformly by `GravitationalWaveTransientLikelihood`, `EquationofStateLikelihood`, `EMTransientLikelihoo` — not a leaky design. But `MultiMessengerLikelihood.setup_from_args` (`nmma/joint/joint_likelihood.py:89-177` is a 90-line procedural dispatcher that self-documents its own fragility inline (`# FIXME: Find a better way to check this automatically`). And configuration handling mixes four serialization mechanisms end-to-end (CLI parsers, pickled data dumps between `nmma-generation`/`nmma-analysis` with **no schema/version check**, YAML, HDF5) — the unversioned pickle dump is the most fragile as an inter-process contract: a stale dump from before a code change can silently misbehave.

2 Prioritized action items (retreat punch list)

P0 — do first (small effort, high leverage, unblocks everything else)

1. Stand up **coverage-gated CI** for `eos/`, `gw/`, `mlmodel/`, `population/`, `post_processing/`, and `core/mpi_setup.py` — even one smoke test per module raises the floor. (Ties to #196, #299.)
2. Delete or replace `nmma/tests/models.py` — a fully-skipped test file is worse than no file.
3. Fix the two concrete latent bugs: the `base.py` `constraints` setter (`UnboundLocalError` → clear `ValueError`) and `NeutronStarPopulation.__init__` (guard unknown `model_name`). Both are small, mechanical, and easy to unit-test alongside the fix.
4. Replace the 11 bare `except:` clauses → catch specific exceptions or at minimum `except Exception:` with a logged message.
5. Fix both broken doc entry points: `doc/index.rst`'s Installation section and `doc/lfi_analysis.md:20` (replace with `pip install -e ".[dev,grb,neuralnet]"`), and `doc/quick-start-guide.rst`'s example command (bad flags → real, working flags). These are the very first things a new user hits — highest leverage-per-minute fix in the whole review.
6. Bump `pre-commit` tool pins (`black`, `pre-commit-hooks`) and pin GitHub Actions versions consistently; replace `actions/checkout@main` in `publish-to-pypi.yml` with a pinned version — a moving-branch checkout in a *publish* workflow is a real supply-chain risk.
7. Repo hygiene sweep: remove stray editor files, delete confirmed-dead `loadEventSpec()`, fix the un-prefixed regex in `core/utils.py:129`, add a `.gitattributes` binary-diff rule to stop future mega-diff PRs.

P1 — do this week (medium effort)

8. Triage and resolve the ~50–65 `FIXME`/`TODO` comments, prioritizing the two densest clusters (`joint/injection_handling.p` `em/model.py`).
9. Sweep the 106 stray `print()` calls into existing loggers.
10. Close or rebase the two mega-diff stale PRs (#105, #82).
11. Resolve version-key duplication: `distance` vs `luminosity_distance` (#375).
12. Decide and document a **versioning/release policy** (#183, #429, #184) — blocked partly by #335 and #382.
13. Reconcile CLI docs against `[project.scripts]` (18 entry points, 7 never invoked in any test); fix `doc/models.md`'s stale line anchor.
14. Decide the long-term status of `nmma/mlmodel/` (maintain-with-tests vs. explicitly deprecate).

P2 — track for next cycle (larger, needs design time)

15. Resolve `core_model_name` design ambiguity (#388).
16. Decouple training/analysis code from the pinned TensorFlow version ceiling (#335, #334), and address poor training results on the new grid (#301).
17. Redesign `gwem_resampling`'s KDE/approximation scheme (#78).
18. Add a schema/version check to the `*_dump.pickle` inter-process contract between `nmma-generation` and `nmma-analysis`.
19. Consider adding a `conda`-first `environment.yml` (#206).
20. Investigate the long-standing dynesty/ultranest indefinite-hang bug (#202).

3 Split by who should do it

3.1 Needs human/expert judgment (physics, API design, deprecation calls)

These require domain expertise, a design decision, or the authority to break/deprecate an API — not appropriate to hand to an AI agent unsupervised.

- Resolve the physics FIXMEs in `nmma/core/conversion.py` (BNS vs NSBH GRB ejecta fitting, prompt-collapse thresholds, `dynamic_mass_fitting_KrFo` routing) — scientific correctness questions, not code bugs.
- `nmma/em/lightcurve_generation.py:145` — whether the $(1+z)$ vs $(1+z)^2$ bolometric luminosity correction is right.
- Decide the fate of `core_model_name` (#388).
- Decide the long-term status of `nmma/mlmodel/` (maintain vs. deprecate).
- Redesign or explicitly accept `MultiMessengerLikelihood.setup_from_args` as the ad-hoc dispatcher it currently is.
- Decide on a schema/versioning strategy for the `*_dump.pickle` inter-process contract.
- Versioning/release policy (#183, #429, #184) and what "NMMA 1.1" should contain.
- TensorFlow version ceiling (#335) and training-quality regression (#301, #302) — require actually re-validating training code, a nontrivial numerical-behavior check.
- Redesign of `gwem_resampling`'s KDE scheme (#78) and the sequential Bayesian Hubble-constant update (#100) — statistical methodology decisions.
- Deciding what to do with the two mega-diff stale PRs (#105, #82) and the 2023-era draft PR (#215).
- Deciding which of the ∼50–65 FIXMEs represent real physics bugs vs. acceptable approximations.
- Coverage strategy for `eos//gw//mlmodel// mpi_setup.py` — what counts as a meaningful test needs a domain expert.
- Population module correctness (`NeutronStarPopulation` only supports 2 hardcoded distributions) — generalizing is a design call.
- The `richpool` vs `schwimmbad` question (#432, PR #433) — touches the untested `mpi_setup.py`; needs MPI validation expertise.
- The long-standing dynesty/ultranest indefinite-hang bug (#202) — deep sampler-specific debugging.
- Milky-Way extinction law (G23) + SFD recalibration (#426) — a physics-correctness change to shipped results, needs expert sign-off.
- Reviewing/merging the three large, physics-adjacent open PRs: #409 (RA/Dec units fix, correctness-critical), #397 (tensor-loading refactor), #379 (new LANL1D model, now with merge conflicts, needs science review).

3.2 AI-agent-tractable (mechanical, well-specified, easy to review)

These are good candidates to hand to an AI coding agent, with human review of the (small) diff.

- Fix both broken doc entry points (install instructions; quick-start-guide's bad flags) — purely mechanical, verifiable by running the corrected commands.
- Add the missing `ValueError` in `NeutronStarPopulation.__init__`; fix the `UnboundLocalError` in `base.py`'s `constraints` setter.
- Replace the 11 bare `except:` with `except Exception:` + logging.
- Sweep the 106 stray `print()` calls into existing loggers.
- Delete/replace the fully-skipped `nmma/tests/models.py`.
- Remove stray editor files; delete confirmed-dead `loadEventSpec()`; fix the un-prefixed regex in `core/utils.py:129`; add a `.gitattributes` binary-diff rule.
- Bump `black`/`pre-commit-hooks` pins and pin GitHub Actions versions consistently.
- Write smoke tests for the untested pure-ish functions in `post_processing/`, `eos/eos_processing.py`, and `core/mpi_setup.py`. Directly matches **#196**/**#299** — good agent-assignable tasks once a human specifies expected test cases.
- Fix arm64 install docs per the working recipe already given in **#374**.
- **#375** deprecated-key cleanup — the issue body already lists the exact files; final "which key wins" call is P1 human sign-off.

- **#382** `Bu2022mv` model not found — narrow, reproducible bug report with a code snippet already in the issue.
- **#357** timeshift dropped by `processData()` — bounded bug, clear failure mode.
- **#263** DOI-fetch 429 errors — cache the Zenodo DOI lookup once instead of hitting the API from every MPI rank.
- **#81** "sample times double call" — remove the redundant `sample_times` argument; redo small rather than resurrect the abandoned PR #82.
- Small doc cross-reference sweep: confirm every `[project.scripts]` entry point has a doc mention; fix `doc/models.md`'s stale line anchor.
- Mechanical first pass at missing docstrings in `nmma/core/base.py` and `nmma/joint/joint_likelihood.py`.
- Additional narrowly-scoped issues with clear specs: **#206** conda `environment.yml`, **#346** JSON lightcurves in example files, **#307** joint-analysis docs, **#136** Zenodo quickstart docs, **#181** compatibility list, **#188** `-debug` CLI flag, **#142** plot multiple models (`good first issue`, reference implementation linked), **#184** release-notes writing (once policy is set).
- PR **#276** (apptainer definition file) — small, self-contained, already sanity-checked; just needs a rebase/re-verify and merge.

4 Detailed audit

4.1 Test coverage gaps

- **`nmma/gw/` has no dedicated test file.** `gw_likelihood.py` (247 lines), `gw_inputs.py` (35 lines — its own header calls it "a hacky subclass to adapt bilby_pipe data generation"), and `gw_parsing.py` are exercised only indirectly through `nmma/tests/joint_analysis_pipeline.py`.
- **`nmma/eos/` has no dedicated test file.** `eos_likelihood.py` (730 lines, ∼12 classes) is only touched incidentally via joint-pipeline imports. `eos_gen.py` (408 lines), `eos_processing.py` (454 lines), `tov.py` (109 lines) have zero test references.
- **`nmma/mlmodel/`** (1,333 lines across 5 modules + 2 binary weight files) has zero tests and isn't imported by any test file.
- **`nmma/population/`** has zero tests despite being wired directly into `MultiMessengerLikelihood.setup_from_args` and reachable via `-population-model` — and it has a real, untested bug (§1.3).
- **`nmma/core/mpi_setup.py`** (683 lines) has no test coverage at all.
- **`nmma/core/conversion.py`** (1,018 lines) is only exercised indirectly via `nmma/tests/injections.py` — no direct unit tests of `cosmology_to_distance`, `nsbh_parameter_conversion`, `bns_ejecta_conversion`, `grb_energy_conversion`, `remnant_disk_mass_fitting`.
- **CLI entry points never invoked in any test**: `lightcurve-injection-slurm-setup`, `create-lightcurve-slurm`, `svdmodel-download`, `multi-config-analysis`, `plot-svdmodel-benchmarks`, `gwem-Hubble-estimate`, `gwem-resampling`.
- Coverage is heavily skewed toward `injections.py` (693 test lines) and `systematics.py` (24 test functions), while `eos` (1,775 lines) and `gw` (311 lines) get essentially none and `mlmodel` (1,333 lines) gets none at all.

4.2 Code quality / design smells

- **11 bare `except:` clauses** — full list in §1.5.
- **106 `print()` calls in production code**, inconsistently mixed with real logging.
- **Large, multi-responsibility files:** `em/model.py` (1,720 lines), `em/utils.py` (1,244 lines), `core/conversion.py` (1,018 lines), `eos/eos_likelihood.py` (730 lines). A live `flake8 -max-complexity 15` run would very likely flag these four.
- **Binary model weights committed directly to git**, not via LFS: `frozen-flow-weights.pth` (8.0 MB), `similarity_embedding_weights.pth` (1.6 MB).
- No Python-2-isms found on spot check — staleness is concentrated in `mlmodel`/`population`/docs/CI

config, not core language usage.

4.3 Staleness

- **`nmma/mlmodel/*`** last touched 2024-07-18 (`inference.py` 2025-01-09) — over a year stale.
- **`nmma/population/pop_likelihood.py`** last touched 2026-05-04 — noticeably older than sibling files, consistent with being under-maintained.
- **`nmma/em/systematics.py`** last touched 2026-02-03, months behind `em/model.py` (2026-07-04) and `em/training.py` (2026-06-30).
- CI action-version inconsistency and stale pre-commit pins — see §1.8.

4.4 Design-principle concerns

- **Positive finding, high confidence:** the core abstraction is genuinely well-factored (§1.11) — `NMMALikelihoodMixin`/`NMMALikelihood` cleanly define a contract that all domain likelihoods extend uniformly; `MultiMessengerLikelihood` composes them without messenger-specific internals beyond routing by `isinstance`. Not a leaky design.
- `MultiMessengerLikelihood.setup_from_args` is the single most important function in the joint pipeline and it reads as ad hoc (§1.11).
- Four different serialization mechanisms are used end-to-end for configuration (CLI parsers, unversioned pickle dumps, YAML, HDF5) — the pickle dump is the most fragile as an inter-process contract.
- `nmma/em/__init__.py` eagerly imports the entire submodule tree, pulling in `healpy` and other heavy dependencies for any small import — avoidable coupling, and the direct cause of the test-collection failures in §1.2.

5 GitHub issue triage (36 open, as of 2026-08-23)

#	Title	Age	Class.	Rationale
432	Using richpool	2026-08	Needs human	Dependency-swap proposal for untested `mpi_setup.py`; paired with PR #433.
429	NMMA 1.1 release?	2026-07	Needs human	Release/admin-rights process decision.
426	Milky-Way extinction law (G23) + SFD recalibration	2026-07	Human (AI can draft)	Physics-correctness change to shipped results.
400	`default_prior` in `InjectionCreator`	2025-08	Needs human	Blocked on upstream bilby/bilby_pipe design decision.
388	Remove reliance on `core_model_name`	2024-11	Needs human	API/design change to SVD model naming/loading.
382	`Bu2022mv` not found in models list	2024-08	AI-tractable	Clear repro snippet already in the issue.
375	Change deprecated key `distance`→`luminosity_distance`	2024-08	AI-tractable	Issue enumerates every file to change.
374	Update docs for arm64	2024-08	AI-tractable	Fix already given in the issue.
357	Timeshift dropped by `processData()`	2024-04	AI-tractable	Bounded bug, clear failure mode.
346	Add JSON lightcurves to example files	2024-03	AI-tractable	Simple additive task.
335	Relax tensorflow version requirement	2024-03	Needs human	Requires re-validating training against newer TF.
307	Joint analysis documentation	2024-01	AI-tractable	Pure doc-writing task.
302	Auto-train tensorflow models on updated grids	2024-01	Needs human	New automation/infra feature.
301	Poor tensorflow training results for new grid	2024-01	Needs human	ML/training-quality diagnosis.
299	Joint analysis unit test	2024-01	AI-tractable	Matches a test-coverage gap in this audit.
263	DOI not retrievable with too many processes	2023-10	AI-tractable	Concurrency bug, well-understood fix.

#	Title	Age	Class.	Rationale
257	Calculate chi2 between observation and grid	2023-10	Needs human	New feature, needs design decision on statistic.
213	Zenodo initialization without an analysis	2023-08	Needs human	API-design decision.
206	`environment.yml` for conda	2023-08	AI-tractable	Simple, self-contained.
203	E(B-V) contribution in likelihood.py	2023-08	Needs human	Physics-modeling feature.
202	dynesty/ultranest samplers hang indefinitely	2023-08	Needs human	Long-standing, hard sampler-hang bug.
196	Tests for `nmma/eos`	2023-08	AI-tractable	Matches this audit's #1 coverage gap.
188	debug flag	2023-08	AI-tractable	Small, scoped CLI feature.
184	Release Notes for Version changes	2023-07	AI-tractable	Doc/process writing (once policy set).
183	Versioning schema	2023-07	Needs human	Policy decision.
181	Add compatibility list to documentation	2023-07	AI-tractable	Doc addition.
159	Move `best_fit.json` into result JSON	2023-07	Stale (`wontfix`)	Already tagged; needs closing.
142	Plot multiple models onto fit lightcurve	2023-06	AI-tractable	`good first issue`, reference linked.
139	Strategy for adding models	2023-06	Needs human	Process/governance discussion.
136	Zenodo quickstart docs	2023-06	AI-tractable	Doc-writing task.
125	`svdmodel_benchmark` multiprocessing bug	2023-06	Stale, triage	3+ years old, unclear relevance.
100	Sequential Bayesian update for Hubble constant	2023-06	Needs human	Statistical-method design question.
83	Additional filter enforcement	2023-05	Stale, triage	Old, thin description.
81	Sample times double call	2023-04	Needs human	Abandoned PR #82 attempt; needs maintainer decision on approach.
78	Improve `gwem_resampling`	2023-03	Needs human	Vague scope, needs statistical design decision.
20	Sampler parameter tuning	2022-04	Stale, triage	Very old, likely superseded by #202.

Issue summary: 15 AI-tractable · 17 needs-human · 4 stale/needs-triage-first (including 1 already `wontfix`).

6 GitHub PR triage (8 open, as of 2026-08-23)

PR	Title	Size	Status	Recommendation
#433	use richpool interface	tiny (+5/-5)	mergeable, review required	Touches untested `mpi_setup.py`; author asks for help validating it works. **Needs human.**
#409	Fix radec units	large (69 files, +2366/-1539)	conflicting, review required	Physics-critical correctness fix, wide blast radius, needs rebase. **Needs human.**
#397	Generalizing Light Curve to Tensor Loading	medium (5 files, +596/-395)	conflicting, stale since Jul 2025	Substantial refactor, needs design opinion and conflict resolution. **Needs human.**
#379	LANL1D model addition	small (+37/-2)	conflicting, previously approved	New physics model; needs science review, has drifted into conflicts since approval. **Needs human** (not the pure fast win it once looked like).

PR	Title	Size	Status	Recommendation
#276	add apptainer definition file	tiny (+25/-0)	mergeable, review required	Small, self-contained, already sanity-checked by author. **AI-tractable / fast win.**
#215	multi_model_analysis command	draft, conflicting, small (+29/-0)	stale since Sep 2023	Needs design review of interaction with existing multi-model support. **Needs human.**
#105	Remove old Bu2019lm pickle / install updates	draft, conflicting, **+3.38M/-4.8k lines, 222 files**	stale since Jun 2023	Binary/pickle files likely diffed as text; needs a maintainer to assess before review is even possible. **Needs human — close or rebase from scratch.**
#82	removed sample_times arg in generate_lightcurve	conflicting, +3.37M/-2.6k lines, 205 files	stale since Jun 2023	Same red flag as #105; not all call sites validated. Maps to issue **#81** — intent still valid, worth redoing small. **Needs human to decide close/redo; the redo itself is AI-tractable.**

PR summary: 1 AI-tractable (fast win) · 7 needs-human. The PR queue skews heavily toward "needs human" because the small/easy ones get merged quickly — what's left is either physics-critical, has merge conflicts, or both.

7 Summary counts

Category	Count
Headline findings	11
P0 / P1 / P2 action items	7 / 7 / 6
Needs human/expert judgment (§3.1)	17
AI-agent-tractable (§3.2)	∼20
Open issues — AI-tractable	15
Open issues — needs human	17
Open issues — stale/needs triage	4
Open PRs — AI-tractable	1
Open PRs — needs human	7

Bottom line for the retreat: the core likelihood abstraction (`NMMALikelihoodMixin`/`MultiMessengerLikelihood`) is sound and doesn't need architectural surgery. The real risk is concentrated in four places: (1) four physics-critical/infra modules (`gw`, `eos`, `mlmodel`, `mpi_setup.py`) with little-to-no test coverage, (2) two shipping-but-untested bugs plus a ∼50–65-item FIXME backlog encoding real open physics questions, (3) two actively-broken doc entry points that fail a new user on their very first command, and (4) a backlog of ∼20 stale-to-old issues and several multi-hundred-file PRs with merge conflicts that need a maintainer triage pass before anyone — human or AI — can act on them productively. The AI-tractable lists above are all good candidates to hand to coding agents *during* the retreat, freeing human time for the architecture/physics/triage decisions that only the team can make.

8 Suggested retreat schedule sketch

- **Day 1 AM**: Triage session — walk through §5/§6 as a group, assign owners, close what's dead (#159, re-triage #125/#83/#20), merge the fast win (#276), decide what to do with #379/#409/#397 given their merge-conflict state.
- **Day 1 PM – Day 2**: P0 items (§2) — coverage scaffolding for `eos/gw/mlmodel/population/mpi_setup.py`, the two concrete bug fixes, bare-except and doc fixes, tooling/CI-pin bumps. Good pairing exercise: hu-

man defines what a correct test looks like, agent writes the boilerplate. Run `flake8 -max-complexity 15` for real on day 1 to corroborate §4.2.

- **Day 3**: Physics/design discussions from §3.1 — `conversion.py` FIXMEs, the `mlmodel/` maintain-vs-deprecate call, `core_model_name`, the pickle-dump versioning question, versioning/release policy. This is the block that most needs everyone in the room together.
- **Day 4**: Work through P1 items and the agent-tractable issue list (§3.2/§5) in parallel — pair each person with an agent session tackling one mechanical item at a time, human reviews before merge.
- **Day 5**: Wrap-up — decide the 1.1 release scope (#429) now that the blockers above are visible, write release notes/changelog policy (#183/#184), retro on what worked.